

Deep Generative Models

Lecture 9

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AI Masters

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Recap of Previous Lecture

ELBO of the Gaussian Diffusion Model

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) - \\ - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t}$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}),$$

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}_{\theta, t}(\mathbf{x}_t), \boldsymbol{\sigma}_{\theta, t}^2(\mathbf{x}_t))$$

It is assumed that $\boldsymbol{\sigma}_{\theta, t}^2(\mathbf{x}_t) = \tilde{\beta}_t \mathbf{I}$.

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta, t}(\mathbf{x}_t)\|^2 \right]$$

Recap of Previous Lecture

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

Reparameterization

$$\begin{aligned}\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \epsilon \\ \mu_{\theta,t}(\mathbf{x}_t) &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \epsilon_{\theta,t}(\mathbf{x}_t)\end{aligned}$$

$$\mathcal{L}_t = \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[\frac{(1 - \alpha_t)^2}{2\tilde{\beta}_t\alpha_t(1 - \bar{\alpha}_t)} \left\| \epsilon - \epsilon_{\theta,t}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon) \right\|^2 \right]$$

At each step in the reverse diffusion process, our goal is to predict the noise ϵ added by the forward process!

Simplified Objective

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t \sim U\{2, T\}} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left\| \epsilon - \epsilon_{\theta,t}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon) \right\|^2$$

Recap of Previous Lecture

Training DDPM

1. Draw a sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$.
2. Sample a timestep $t \sim U\{1, T\}$ and noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.
3. Obtain the noisy image: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$.
4. Compute the loss: $\mathcal{L}_{\text{simple}} = \|\epsilon - \epsilon_{\theta,t}(\mathbf{x}_t)\|^2$.

Sampling with DDPM

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute the mean of $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \sigma_t^2 \cdot \mathbf{I})$:

$$\boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \epsilon_{\theta,t}(\mathbf{x}_t)$$

3. Generate a denoised image: $\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) + \sigma_t \cdot \epsilon$, where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.

Outline

1. Continuous-Time Normalizing Flows

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Discrete-Time Normalizing Flows

Change of Variable Theorem (CoV)

Let $\mathbf{x}$ be a random variable with density $p(\mathbf{x})$, and let $\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a differentiable and **invertible** transformation. If $\mathbf{z} = \mathbf{f}(\mathbf{x})$, $\mathbf{x} = \mathbf{f}^{-1}(\mathbf{z}) = \mathbf{g}(\mathbf{z})$, then

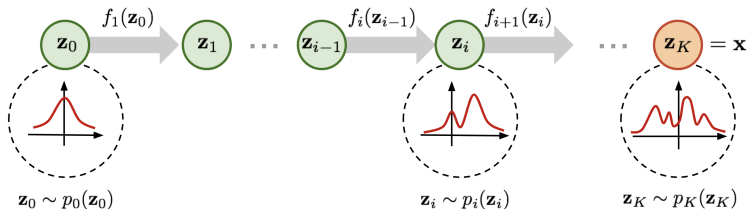
$$p(\mathbf{x}) = p(\mathbf{z}) |\det(\mathbf{J}_{\mathbf{f}})| = p(\mathbf{z}) \left| \det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) \right| = p(\mathbf{f}(\mathbf{x})) \left| \det \left(\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}} \right) \right|$$

Discrete-Time Normalizing Flows

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$$\log p_{\theta}(\mathbf{x}) = \log p(\mathbf{f}_K \circ \dots \circ \mathbf{f}_1(\mathbf{x})) + \sum_{k=1}^K \log \left| \det \left(\frac{\partial \mathbf{f}_k}{\partial \mathbf{f}_{k-1}} \right) \right|.$$

Towards Continuous-Time Normalizing Flows

Up to this point, we have considered discrete-time normalizing flows:

$$\mathbf{x}_{t+1} = \mathbf{f}_{\theta}(\mathbf{x}_t, t); \quad \log p(\mathbf{x}_{t+1}) = \log p(\mathbf{x}_t) - \log \left| \det \frac{\partial \mathbf{f}_{\theta}(\mathbf{x}_t)}{\partial \mathbf{x}_t} \right|.$$

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Residual Flows

Let's consider the flow $\mathbf{f}_{\theta}(\mathbf{x}, t) = \mathbf{x} + \mathbf{v}_{\theta}(\mathbf{x}, t)$:

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- ▶ Here $\mathbf{f}_{\theta}$ is the NF **bijection**, $\mathbf{v}_{\theta}$ is the **velocity** (vector field).
- ▶ The update $\mathbf{x}_{t+1} - \mathbf{x}_t = \mathbf{v}_{\theta}(\mathbf{x}_t, t)$ is a **finite-difference** approximation of a derivative.

Towards Continuous-Time Normalizing Flows

Residual dynamics $\mathbf{x}_{t+1} = \mathbf{x}_t + \mathbf{v}_\theta(\mathbf{x}_t, t)$ is an Euler step with $h = 1$:

$$\frac{\mathbf{x}(t+h) - \mathbf{x}(t)}{h} = \mathbf{v}_\theta(\mathbf{x}(t), t)$$

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Taking the limit $h \rightarrow 0$, we obtain continuous-time dynamics.

Continuous-Time Dynamics

Consider an Ordinary Differential Equation (ODE):

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_\theta(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0.$$

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$$\mathbf{x}(t_1) = \int_{t_0}^{t_1} \mathbf{v}_\theta(\mathbf{x}(t), t) dt + \mathbf{x}_0$$

Here, $\mathbf{v}_\theta : \mathbb{R}^m \times [t_0, t_1] \rightarrow \mathbb{R}^m$ is a **velocity vector field**.

Ordinary Differential Equations (ODEs)

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Flow

Let call **the flow** $\psi : \mathbb{R}^m \times [t_0, t_1] \rightarrow \mathbb{R}^m$ the solution of ODE:

$$\frac{d\psi_t(\mathbf{x}_0)}{dt} = \mathbf{v}_\theta(\psi_t(\mathbf{x}_0), t); \quad \text{with initial condition } \psi_0(\mathbf{x}_0) = \mathbf{x}_0.$$

Numerical Solution of ODEs

$$\psi_t(\mathbf{x}_0) = \int_{t_0}^{t_1} \mathbf{v}_\theta(\mathbf{x}(t), t) dt + \mathbf{x}_0$$

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$$\psi_t(\mathbf{x}_0) = \int_{t_0}^{t_1} \mathbf{v}_{\theta}(\mathbf{x}(t), t) dt + \mathbf{x}_0 \approx \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0, t_1).$$

Here, we require the numerical routine $\text{ODESolve}_v(\mathbf{x}_0, \theta, t_0, t_1)$.

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$\text{ODESolve}_v(\mathbf{x}_0, \theta, t_0, t_1)$ consists of sequence of iterative update steps.

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Euler Update Step

$$\frac{\mathbf{x}(t+h) - \mathbf{x}(t)}{h} = \mathbf{v}_{\theta}(\mathbf{x}(t), t)$$

$$\mathbf{x}(t+h) = \mathbf{x}(t) + h \cdot \mathbf{v}_{\theta}(\mathbf{x}(t), t)$$

Numerical Solution of ODEs

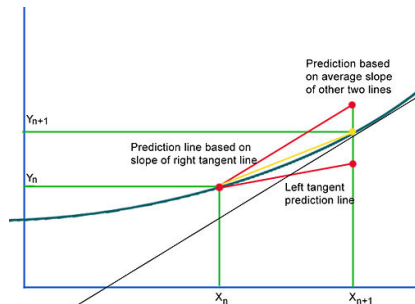
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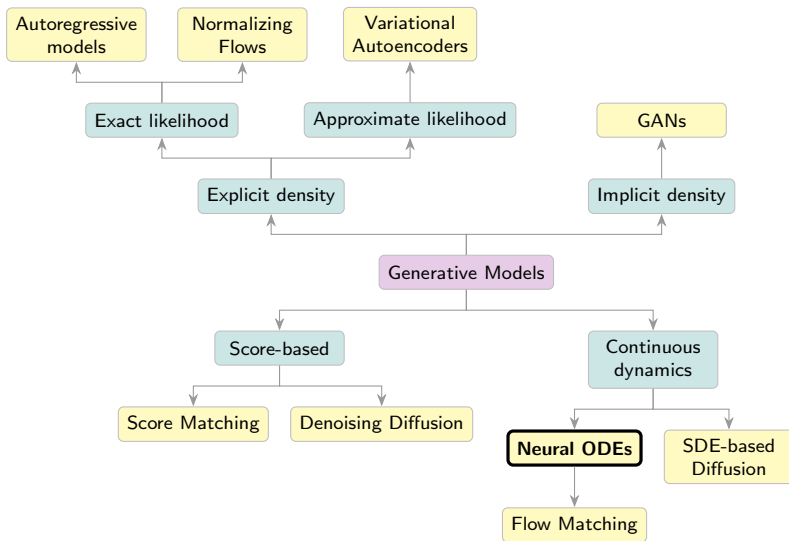
Heun's Update Step

$$\mathbf{x}'(t+h) = \mathbf{x}(t) + h \cdot \mathbf{v}_{\theta}(\mathbf{x}(t), t)$$

$$\mathbf{x}(t+h) = \mathbf{x}(t) + \frac{h}{2} \cdot (\mathbf{v}_{\theta}(\mathbf{x}(t), t) + \mathbf{v}_{\theta}(\mathbf{x}'(t+h), t+h))$$

Image credit: https://en.wikipedia.org/wiki/Heun's_method

Generative Models Taxonomy



Continuous-Time Normalizing Flows: Neural ODE

Neural ODE

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\theta}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0$$

Euler ODESolve

$$\mathbf{x}(t + h) = \mathbf{x}(t) + h \cdot \mathbf{v}_{\theta}(\mathbf{x}(t), t)$$

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- ▶ Consider $[t_0, t_1] = [0, 1]$ for simplicity.
- ▶ If $\mathbf{x}(0)$ is a random variable with density $p_0(\mathbf{x})$,
- ▶ Then, for any t , $\mathbf{x}(t)$ is a random variable with density $p_t(\mathbf{x})$.

Continuous-Time Normalizing Flows: Intuition

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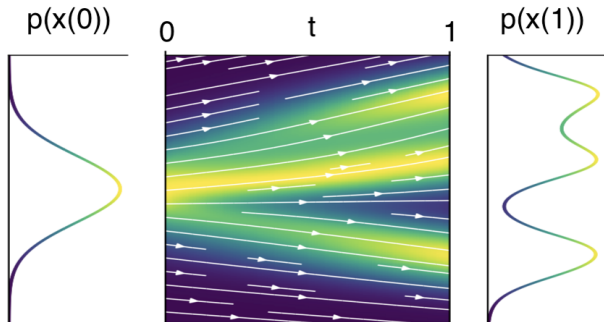
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- ▶ $p_t(\mathbf{x}) = p(\mathbf{x}, t)$ describes the **probability path** interpolating between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.
- ▶ What is the difference between $p_t(\mathbf{x}(t))$ and $p_t(\mathbf{x})$?

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Continuous-Time Normalizing Flows: Reversibility

Theorem (Picard)

If $\mathbf{v}$ is continuously differentiable with a bounded derivative in $\mathbf{x}$ and continuous in t , then the ODE has a **unique solution** given by a flow ψ_t .

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This guarantees the ODE is **uniquely reversible**.

$$\psi_1(\mathbf{x}_0) = \mathbf{x}_0 + \int_0^1 \mathbf{v}_\theta(\psi_t(\mathbf{x}_0), t) dt$$

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Note: Unlike discrete-time flows, $\mathbf{v}$ need not be invertible (uniqueness ensures bijection). How can we compute $p_t(\mathbf{x})$ at arbitrary t ?

Summary

- ▶ Classifier guidance provides a technique to turn an unconditional model into a conditional one by training an auxiliary classifier on noisy data.
- ▶ Classifier-free guidance removes the need for such a classifier, yielding a practical recipe now widely used.
- ▶ Continuous-time normalizing flows leverage neural ODEs to define continuous-time trajectories $\mathbf{x}(t)$, relaxing many constraints of discrete-time flows.
- ▶ If $\mathbf{x}_0$ is a random variable, this yields a **probability path** $p_t(\mathbf{x})$ as time evolves.